Statistics 280 — Practice Midterm 1C

Unofficial practice exam, written in the style of the W25 and W26 Midterm 1.
Covers Part 1: study design, descriptive statistics, the normal curve, Markov & Chebyshev, correlation, regression.

Name _____ Start time / End time _____

- Answer all questions in the space provided and indicate where you have written additional work.
- Solutions without justification will be marked as wrong for all questions except the scatter-plot matching question. Show all work. A calculator command does not count as justification.
- Express answers as fractions or as decimals rounded to 2 decimal places.
- The last page is a table of values from the normal distribution. Closed book, no notes. A calculator that can take square roots is allowed.
- Suggested time: 80 minutes. Total: 200 points.

Question	**1**	**2**	**3**	**4**	**5**	**6**	**Total**
Points	30	30	35	35	30	40	200
Score							

1. (30 points) A drug company tests a new migraine drug. It recruits 1,000 volunteers with ads asking "Are your migraines worse than ever?" The volunteers are randomly assigned: 500 get the drug and 500 get a placebo pill that looks identical. Neither patients nor doctors know who gets what. Average migraine days per month:

Group	Number	Before	After 3 months
Drug: all	500	8.0	5.0
Drug: adherers (took ≥ 80% of pills)	350	8.0	4.2
Drug: non-adherers	150	8.0	6.9
Placebo: all	500	8.0	6.5
Placebo: adherers	340	8.0	5.9
Placebo: non-adherers	160	8.0	7.8

a) (6 points) The placebo group improved from 8.0 to 6.5 days even though the placebo has no active ingredient. Give two reasons. (Hint: think about how the volunteers were recruited.)

b) (6 points) Estimate the effect of the drug. Which comparison did you use, and why?

c) (6 points) A company scientist compares drug adherers (4.2) to the whole placebo group (6.5) and reports an effect of 2.3 days. What is wrong with this? Is comparing drug adherers to placebo adherers (4.2 vs 5.9) better? Is it a perfect comparison?

d) (6 points) Why is it important that the study is double-blind? Give a specific way the results could be biased if the doctors knew which patients got the drug.

e) (6 points) The company's original plan was to give all 1,000 volunteers the drug and compare their migraine days to their own previous months. Using the table, roughly how large an effect would that design have reported, and why would it be misleading?

2. (30 points)

a) (8 points) Find the value of u that minimizes $f(u) = |u-1| + |u-4| + 2|u-6| + 3|u-10|$, and the value that minimizes $g(u) = (u-1)^2 + (u-4)^2 + 2(u-6)^2 + 3(u-10)^2$.

b) (6 points) Find the value of u that minimizes $h(u) = (2u-3)^2 + (2u-7)^2 + (u-5)^2$. (Hint: $(2u-3)^2 = 4(u-1.5)^2$.)

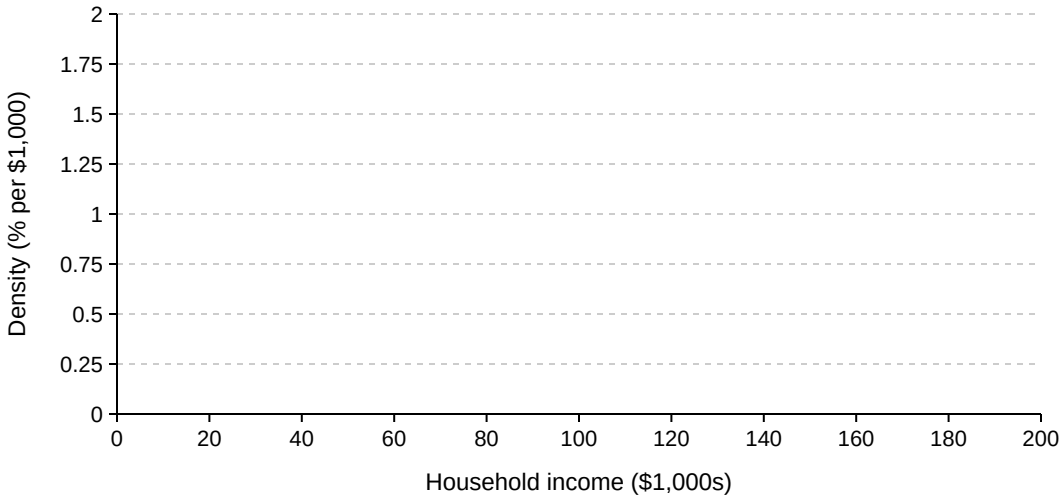
c) (8 points) A list of 20 numbers has mean 50 and SD 10. One entry was recorded as 50 but should have been 90. Find the corrected mean and SD.

d) (3 points) For a list of 5 numbers, R's `sd()` function returns 4. What is the SD as defined in class?

e) (5 points) Using the definition of percentile from class, give a list of 5 numbers whose SD is larger than its IQR, and a list of 5 numbers whose IQR is larger than its SD. Verify both.

3. (35 points) Household incomes in a town (in thousands of dollars) have the following percentiles: 0th: 0; 10th: 20; 30th: 40; 60th: 60; 90th: 100; 100th: 200. Assume incomes are evenly distributed between consecutive listed percentiles.

a) (8 points) Draw the histogram on the density scale, using class intervals whose endpoints are the listed values. Label the heights of the blocks.



b) (8 points) Estimate the median and the mean. Which is larger, and why?

c) (5 points) What percentage of households have incomes between $50,000 and $120,000?

d) (6 points) Estimate the interquartile range.

e) (5 points) A new policy sets each household's after-tax income to $0.8 \times \text{income} + 3$ (in thousands). Find the median, mean, and IQR of after-tax income.

f) (3 points) The mayor wants one number for a "typical household's income." The tax office wants to estimate the town's total income from the number of households. Which measure of center should each use?

4. (35 points) Scores on an aptitude test follow the normal curve with mean 100 and SD 15 (parts a–d).

a) (4 points) Estimate the percentage of test takers who score above 130.

b) (6 points) Estimate the percentage who score between 85 and 120.

c) (6 points) A program admits the top 15% of scorers. Estimate the cutoff score.

d) (6 points) Among admitted students, what percentage scored above 130?

e) (5 points) *Do not assume the normal curve for e) – g).* A list of non-negative numbers has mean 20. Use the reasoning behind Markov's inequality to find the largest possible value of the 75th percentile and of the median.

f) (4 points) For any list, at least what percentage of the values lie within 1.5 SDs of the mean? Compare with the normal curve.

g) (4 points) A list has mean 50 and SD 5. Is it possible that 20% of the values are 62 or more? Explain.

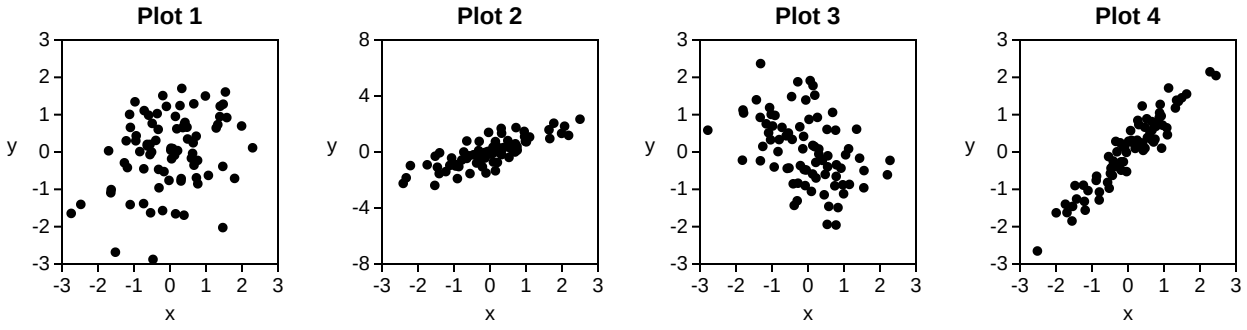
5. (30 points)

a) (14 points) Two groups of students report hours studied per week (x) and test score (y).

Group	n	avg x	SD x	avg y	SD y	r
1	60	10	2	50	5	0.4
2	40	15	2	60	5	0.4

(i) Find the average of xy in each group. (ii) Find the averages and SDs of x and y in the combined data. (iii) Find the correlation in the combined data. Why is it different from 0.4?

b) (6 points) Match the correlations 0.3, 0.8, −0.5, and 0.95 to the scatter plots below. Look carefully at the axes. (No justification needed.)



c) (10 points) True or false? Explain each briefly.

(i) If $r = 0$, then x and y are not associated.

(ii) If every y is multiplied by −2, the correlation is multiplied by −2.

(iii) The correlation between state average education and state average income is likely larger than the correlation between education and income for individuals.

(iv) Among students with SAT scores above 1400, the correlation between SAT and first-year GPA is likely smaller than among all students.

(v) A correlation of 0.8 means that 80% of the points lie on a straight line.

6. (40 points) An ice cream shop records the daily high temperature x (°F) and daily sales y (dollars) over many summer days. Average temperature 75°F, SD 8°F; average sales \$1,200, SD \$300; $r = 0.75$. The scatter plot is football-shaped.

a) (6 points) Find the equation of the regression line for predicting sales from temperature.

b) (5 points) Predict sales on an 87°F day.

c) (8 points) Find the RMS error. On 87°F days, find the range that contains the middle 90% of sales.

d) (7 points) On one 87°F day, sales were \$1,300. Find the residual. Roughly what percentile is \$1,300 among 87°F days? Among all days?

e) (4 points) Would you use the regression line to predict sales on a 20°F day? Explain.

f) (6 points) The temperatures are converted to Celsius, $C = \frac{5}{9}(F - 32)$. Which of the correlation, the slope of the regression line, and the RMS error change? Give their new values.

g) (4 points) True or false? (i) For any data set, the RMS error is at most SD(y). (ii) For any data set, the SD of y within each vertical strip is at most SD(y).

A Normal Table

Height = height of the standard normal curve at z (percent per standard unit). Area = percentage of the area under the normal curve between −z and z.

z	Height	Area	z	Height	Area	z	Height	Area
0.00	39.89	0.00	**1.50**	12.95	86.64	**3.00**	0.443	99.730
0.05	39.84	3.99	**1.55**	12.00	87.89	**3.05**	0.381	99.771
0.10	39.70	7.97	**1.60**	11.09	89.04	**3.10**	0.327	99.806
0.15	39.45	11.92	**1.65**	10.23	90.11	**3.15**	0.279	99.837
0.20	39.10	15.85	**1.70**	9.40	91.09	**3.20**	0.238	99.863
0.25	38.67	19.74	**1.75**	8.63	91.99	**3.25**	0.203	99.885
0.30	38.14	23.58	**1.80**	7.90	92.81	**3.30**	0.172	99.903
0.35	37.52	27.37	**1.85**	7.21	93.57	**3.35**	0.146	99.919
0.40	36.83	31.08	**1.90**	6.56	94.26	**3.40**	0.123	99.933
0.45	36.05	34.73	**1.95**	5.96	94.88	**3.45**	0.104	99.944
0.50	35.21	38.29	**2.00**	5.40	95.45	**3.50**	0.087	99.953
0.55	34.29	41.77	**2.05**	4.88	95.96	**3.55**	0.073	99.961
0.60	33.32	45.15	**2.10**	4.40	96.43	**3.60**	0.061	99.968
0.65	32.30	48.43	**2.15**	3.96	96.84	**3.65**	0.051	99.974
0.70	31.23	51.61	**2.20**	3.55	97.22	**3.70**	0.042	99.978
0.75	30.11	54.67	**2.25**	3.17	97.56	**3.75**	0.035	99.982
0.80	28.97	57.63	**2.30**	2.83	97.86	**3.80**	0.029	99.986
0.85	27.80	60.47	**2.35**	2.52	98.12	**3.85**	0.024	99.988
0.90	26.61	63.19	**2.40**	2.24	98.36	**3.90**	0.020	99.990
0.95	25.41	65.79	**2.45**	1.98	98.57	**3.95**	0.016	99.992
1.00	24.20	68.27	**2.50**	1.75	98.76	**4.00**	0.013	99.9937
1.05	22.99	70.63	**2.55**	1.54	98.92	**4.05**	0.011	99.9949
1.10	21.79	72.87	**2.60**	1.36	99.07	**4.10**	0.009	99.9959
1.15	20.59	74.99	**2.65**	1.19	99.20	**4.15**	0.007	99.9967
1.20	19.42	76.99	**2.70**	1.04	99.31	**4.20**	0.006	99.9973
1.25	18.26	78.87	**2.75**	0.91	99.40	**4.25**	0.005	99.9979
1.30	17.14	80.64	**2.80**	0.79	99.49	**4.30**	0.004	99.9983
1.35	16.04	82.30	**2.85**	0.69	99.56	**4.35**	0.003	99.9986
1.40	14.97	83.85	**2.90**	0.60	99.63	**4.40**	0.002	99.9989
1.45	13.94	85.29	**2.95**	0.51	99.68	**4.45**	0.002	99.9991